A Case of 7-Hydroxymitragynine Use Requiring Inpatient Medically Managed Withdrawal

Rachel S. Wightman, MD and David Hu, MD

Objectives: There is an increased interest in kratom and kratom-related products such as 7-hydroxymitragynine (7-OH). Here we present the case of a patient seeking inpatient substance use treatment for daily 7-OH use and the clinical course of care including clinical opiate withdrawal scale (COWS) and medication treatment response.

Methods: This is a single case report from an inpatient medically managed withdrawal level of substance use treatment care.

Results: A 38-year-old man with a history of opioid use disorder reported escalation of use including kratom to 7-OH complicated by the development of use disorder, tolerance, and withdrawal. The patient experienced a withdrawal syndrome consistent with opioid withdrawal upon stopping use, with COWS peaking at 14. The patient was treated with a short course of buprenorphine sublingual (daily dosing 2–8 mg) based on signs and symptoms. He did not want to remain on buprenorphine after the medically managed withdrawal period and was successfully tapered off buprenorphine during his treatment stay and transitioned to residential level of care.

Conclusions: Given the widespread availability of kratom and kratom-related products it is important for health care providers to screen for use and development of use disorder and be prepared to provide counseling, support, and treatment for patients.

Key Words: buprenorphine, kratom, 7-hydroxymitragynine, kratom use disorder

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Kratom (*Mitragyna speciosa*) is a tree native to Southeast Asia. Leaves are used to make tea, dried and turned into powder, and processed to create additional products including concentrates and semi-synthetics. Kratom alkaloids have shown activity at opioid, serotonergic, dopaminergic, and noradrenergic pathways, and although a clear dose-response relationship has not been established, kratom products are reported to have both stimulant and sedative effects depending on dose.[1]

The FDA has designated kratom as a new dietary ingredient based on inadequate information to suggest that kratom "does not pose a significant or unreasonable risk of illness or injury." As such, any product that contains kratom is considered adulterated under section 402(a)(2)(C)(i).[2] Despite this determination by the FDA, states and municipalities have taken different approaches to kratom use, sale, and regulation. Some states have scheduled mitragynine and 7-hydroxymitragine (7-OH) and some municipalities have banned sales of kratom and related products. Many states have adopted legislation promoted by industry to create a regulatory structure for kratom products which mirrors herbal and dietary supplement regulation in direct conflict with the FDA. Kratom products are available for sale in convenience stores, head shops, and over the internet in states with and without kratom bans.[3]

Recently, there has been increased interest in semi-synthetic kratom alkaloid products containing primarily 7-OH. In kratom leaf preparations, 7-OH presence is minimal, but 7-OH is formed in the body via metabolism of mitragynine, the most abundant alkaloid in kratom, and can also be formed from oxidation of dried plant material. Mitragynine and 7-OH both demonstrate partial agonism at the mu opioid receptor; however, 7-OH demonstrates higher efficacy and is more potent.[4] Previous case reports have documented physiologic dependence, withdrawal upon stopping use, and development of kratom use disorder related to chronic daily use.[5] Here we present, to our knowledge, the first published case of a patient seeking inpatient medically managed withdrawal for treatment of 7-OH use.

METHODS

This is a single case report. The case was identified prospectively and data were extracted manually from the electronic health record. Patient consent was obtained for case publication.

Case Details

CC: "I started using 7 hydroxy kratom a few months ago and it got a lot worse. I want to stop."

Case: A 38-year-old man with a history of opioid use disorder (OUD) presented to treatment seeking

From the Brown Emergency Medicine, Alpert Medical School of Brown University, Providence, RI (RSW); and Zinnia Health, Gemini Behavioral Health, Wellington, FL (DH).

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Send correspondence and reprint requests to Rachel S. Wightman MD, 55 Claverick St, Providence, RI 02903. E-mail: Rachel_wightman@brown.edu.

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medically managed withdrawal from 7-OH. The patient reported a history of oxycodone use starting at age 16. He developed opioid use disorder in his late teens and later stabilized on prescription buprenorphine maintenance at age 28 for 2 years. The patient was able to remain abstinent from opioids for 1 year after stopping buprenorphine maintenance and then started using kratom at age 31. The patient reported starting to use kratom "to get high" and because it helped his mood, gave him energy, and helped with pain. The patient used kratom capsules 4–5 times daily, totaling ~30 g. Three months before presentation, the patient switched to 7-OH tablets to self-treat neck pain and anxiety. The patient reported improvement in his neck pain, but his use quickly escalated and he developed problems related to his use. The patient reported that 7-OH use caused memory and anxiety problems. He experienced difficulty concentrating at work as a store manager if he didn't take 7-OH throughout his day.

Patient reported 7-OH is readily accessible at gas stations and smoke shops near his workplace. His use caused interpersonal problems as he was hiding his use and spending less time with his family and his regular pastimes. Patient reported he was taking larger and more frequent amounts to avoid withdrawal symptoms that occurred on stopping use, which included anxiety, insomnia, rhinorrhea, abdominal discomfort, restlessness, diaphoresis, and chills. The patient reported spending up to $60 daily, using eight 30 mg tablets daily, and taking them every 1 to 2 hours. Patient reported attempts to cut down or quit on his own 4 or 5 times, but he had not remained abstinent for more than 24 hours.

On presentation, the patient reported cravings 8/10 and withdrawal symptoms of anxiety, restless legs, and abdominal discomfort. His last use of 7-OH was on the day of presentation at 2 PM. At intake breathalyzer was negative and point-of-care urine drug testing was negative for all substances other than cannabinoids. Reflex confirmatory testing was negative for delta-9-THC-COOH. Urine mitragynine testing was negative. The patient was admitted at ~10:30 PM to medically managed withdrawal (ASAM level of care 3.7) with a plan to monitor Clinical Opiate Withdrawal Scale (COWS) and Vital Signs every 4 hours while awake. The care plan included initiation of buprenorphine if COWS > 13.

The patient had a 5-day stay in a closed medically managed withdrawal unit. Please see Table 1 for a summary of patient vital signs, COWS scores, and medication administration during the inpatient stay. The COWS score peaked on day 2 and buprenorphine was initiated. The patient received a total of 4 mg of buprenorphine on day 2, 8 mg of buprenorphine on day 3, 6 mg of buprenorphine on day 4, 4 mg of buprenorphine on day 5, and 2 mg of buprenorphine on day 6. Throughout the inpatient stay, comfort medications including clonidine, methocarbamol, hydroxyzine, trazadone, melatonin, and ibuprofen were available. On day 6 the patient stopped buprenorphine and he was transitioned to the residential level of care.

TABLE 1. Clinical Course Including Vital Signs, Medications Administered, and Clinical Opiate Withdrawal Scales

Day and Time	Blood Pressure (mm Hg)	Heart Rate (beats/min)	COWS	Medications	Dose (mg)
Day 1					
22:20	148/81	96	2		
Day 2					
7:34	126/98	86	4	Clonidine	0.1
10:35			8		
10:41				Ibuprofen	600
10:42				Methocarbamol	500
15:30	130/83	82	14		
16:00				Bup/nal HCl	2–0.5
19:45	122/71	77	4		
20:00				Duloxetine	30
21:34				Methocarbamol	500
21:34				Ibuprofen	600
22:00				Bup/nal HCl	2–0.5
Day 3					
7:57	136/83	81	2		
8:00				Bup/nal HCl	2–0.5
12:00				Bup/nal HCl	2–0.5
12:35				Hydroxyzine	50
18:00				Bup/nal HCl	2–0.5
18:55				Hydroxyzine	50
20:00				Duloxetine	30
21:12	124/79	84	6		
21:14				Ibuprofen	600
21:14				Methocarbamol	500
21:57				Melatonin	5
22:00				Bup/nal HCl	2–0.5
22:30				Trazodone	50
Day 4					
7:30				Bup/nal HCl	2–0.5
8:07	112/74	82	3		
14:30				Bup/nal HCl	2–0.5
19:35	117/75	77	1		
20:00				Duloxetine	30
21:30				Bup/nal HCl	2–0.5
21:48				Methocarbamol	500
21:49				Hydroxyzine	50
22:44				Trazodone	50
Day 5					
8:00				Bup/nal HCl	2–0.5
8:36	103/71	78	2		
10:22				Clonidine	0.1
10:22				Hydroxyzine	50
10:24				Ibuprofen	600
10:24				Methocarbamol	500
19:20	99/68	81			
20:00				Duloxetine	30
20:00				Bup/nal HCl	2–0.5
22:12				Trazodone	50
Day 6					
6:48				Ibuprofen	600
6:52				Methocarbamol	500
6:53	105/69	79			
8:00				Bup/nal HCl	2–0.5
11:30			2		
12:34				Clonidine	0.1
19:09	112/76	80			

Bup/nal HCl indicates sublingual buprenorphine/naloxone HCl.

DISCUSSION

This case demonstrates escalation of use from kratom to 7-OH complicated by the development of use

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disorder, tolerance, and withdrawal. The patient transitioned from being fully abstinent from opioids, then started using kratom for psychoactive effects, and in time transitioned to 7-OH for anxiety and pain. He met criteria for substance use disorder using the DSM-V checklist adapted for 7-OH, including physiologic dependence (tolerance and withdrawal) and criteria within the domains of impaired control and social dysfunction with continued use despite interpersonal problems and problems at work.

Real-world detoxification cases including vital signs and COWS scores before and after pharmacotherapy, to our knowledge, have not been previously described for 7-OH. This patient tried unsuccessfully to quit use on his own on multiple occasions before seeking medically managed withdrawal (detox) for 7-OH. This individual was admitted to inpatient medically managed withdrawal treatment as the case was billed under an opioid use disorder diagnostic code. Other patients requesting inpatient medically managed withdrawal have been pushed to the residential level of care or turned away due to inability to obtain insurance prior authorization and reimbursement for admissions related to kratom or 7-OH. In addition, questions about kratom use and more recently available semi-synthesized kratom products such as 7-OH mitragynine and mitragynine pseudoindoxyl are not routinely asked as part of standard intake and are not tested for via standard urine drug screen panels.

In this case, confirmatory testing was only sent for mitragynine and not 7-OH. As the kratom alkaloidal product market expands to include semi-synthetic or other derived products, testing panels and libraries will need to be expanded. In addition, it has been reported that laboratory testing for 7-OH is more challenging than testing for mitragynine due to a variety of factors.[6] Knowledge of kratom alkaloidal testing limitations and patient product use history is critical to the correct interpretation of testing results for kratom-related product exposure. Further, in this case, the patient had a positive point-of-care test for cannabinoids, but confirmatory testing was negative for delta-9-THC metabolites; this could suggest use of alternative cannabinoids (eg, delta-8-THC) that may cross-react with the carboxy-THC immunoassay.[7]

During the treatment course the patient experienced withdrawal symptoms captured via standardized COWS assessments consistent with opioid withdrawal. Previous case reports and series have documented kratom use disorder and successful treatment with varied courses of buprenorphine.[8–11] While maintenance daily doses of buprenorphine have generally increased in the age of fentanyl, for this individual, smaller doses of buprenorphine over 5 days were effective with adjunctive medications to control withdrawal symptoms. It is important for providers to dose buprenorphine to clinical effect as guidelines do not exist for kratom or 7-OH products. Another option that could be considered would be a single dose of extended-release injectable buprenorphine. However, this approach may be complicated by cost and logistical challenges related to medication ordering and delivery, off-label use, and prior authorization requirements. The patient described did not want to remain on buprenorphine after the medically managed withdrawal period and was successfully tapered off buprenorphine during his treatment stay. The duration of buprenorphine treatment should be patient-centered and consider the patient's history of substance use disorder, symptoms, and goals and desires.

Researchers have previously highlighted concerns about 7-OH products and concentrated kratom formulations. Given the widespread availability of these products it is important for health care providers to understand the risks of kratom and kratom-derived products, to screen for use and development of use disorder, and be prepared to provide counseling, support, and treatment for patients.

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